Learned Manifolds for Digital Personality

Method selection, implementation pathway and empirical rationale

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Decision. Adopt a learned low-dimensional nonlinear encoder-decoder representation, equipped with a decoder-induced Riemannian metric. Use a 16-dimensional latent space as a provisional implementation setting, not as an established intrinsic dimension of personality.

1. Research question and intended contribution

The question is how to use a manifold to form digital personality. Here, digital personality means a persistent, evidence-grounded computational representation that can condition an artificial agent's responses and behavior. It is not a claim that the agent possesses human consciousness, nor that a questionnaire alone captures an entire person.

The proposed answer has two layers: first learn a compact representation from personality-relevant observations; then attach geometry and a behavioral interface to that representation. The present experiment implements and evaluates the first layer and parts of its geometry. Training a personality-conditioned agent, learning longitudinal transitions, and validating behavioral fidelity remain future work.

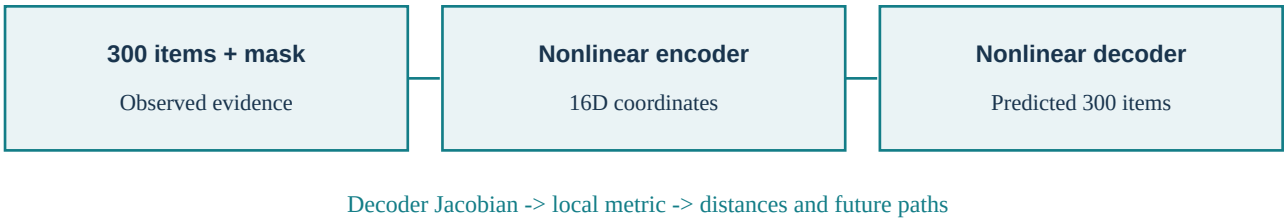


Figure 1. Selected representation pipeline. The manifold is the decoder-defined surface in observation space; the 16D vector is its coordinate representation. Geometry is derived from how changes in coordinates change predicted observations.

Why this decision is not simply a leaderboard choice

A raw-feature ridge baseline currently predicts held-out questionnaire items better. The selected method is preferred for its combined representation, reconstruction and geometry interface, not because it has demonstrated universally superior accuracy. A complete digital personality system must additionally specify how the representation affects behavior and how new evidence changes it.

2. Why choose this model?

High-dimensional evidence, low-dimensional representation

The current input consists of 300 personality questionnaire items rather than only five aggregate scores. This preserves item-level variation that a five-score representation discards. Compression makes the representation reusable by downstream models while retaining an explicit relationship to the original observations. More columns are not automatically better: relevance, coverage and measurement quality matter as much as dimensionality.

Nonlinearity without abandoning observable evidence

A nonlinear encoder and decoder can represent interactions that a linear projection cannot express. This is a capacity argument, not proof that the true psychological structure is nonlinear. The decoder provides an important accountability mechanism: a coordinate change can be inspected through changes in predicted item responses. Coordinates are nevertheless not automatically identifiable traits, and decoder sensitivity is not a causal explanation.

A concrete meaning for manifold geometry

Ordinary Euclidean latent distance treats every coordinate direction equally. A decoder-derived metric instead measures local movement by its effect on reconstructed observations. This links geometric operations to the measurement space. Its psychological meaning still depends on item scaling, coverage and decoder fidelity; standardized item changes are not automatically equivalent to meaningful personality changes.

Requirement	Selected mechanism	Important qualification
Compact individual representation	Amortized nonlinear encoder	A response case is not necessarily a verified unique individual.
Grounding in measurements	Item-level decoder and masked reconstruction	Reconstruction does not establish behavioral validity.
Operational geometry	Jacobian-based local metric	Metric availability does not prove useful curvature.
Integration with future agents	Coordinates condition a generator or policy	No personality-conditioned LLM has yet been validated.
Updating from new evidence	Re-encode a revised observation set	Temporal update dynamics have not yet been learned.

Selection rationale: this model is a coherent engineering foundation for the requested representation-plus-geometry system. It is not yet established as the best psychological model or the best predictive model.

3. Literature foundation and our adaptation

The method combines established components rather than reproducing one paper unchanged. The following comparisons concern mechanisms and research roles, not directly comparable numerical benchmarks across unrelated datasets.

Literature / approach	What it contributes	How the present method differs
Isomap [1]	Neighborhood shortest paths approximate manifold distances for nonlinear embedding.	We learn an encoder and decoder rather than using graph embedding as the primary representation. Isomap remains a useful geometric baseline.
Auto-Encoding Variational Bayes [2]	Amortized inference and reparameterized training of a probabilistic latent model.	We use masked personality-item reconstruction and split-form training. Gaussian posterior regularization is an implementation option, not the definition of a manifold.
Latent Space Oddity [3]	A nonlinear generative decoder can induce local Riemannian geometry; uncertainty matters for its stochastic formulation.	Our current metric uses the deterministic decoder mean plus a small ridge. It does not reproduce the full stochastic metric or its uncertainty treatment.
Traits as distributions of states [4]	Personality theory can accommodate both stable tendencies and within-person variation.	This motivates a future trait/state extension; a same-occasion questionnaire split cannot test that extension.
IPIP public-domain inventory materials [5]	Item-level measurement and official scoring information.	These supply observations and scoring, not a ready-made manifold or digital agent.

What is borrowed, and what is project-specific?

Borrowed foundations are nonlinear latent inference [2] and decoder-induced geometry [3]. The project-specific integration is to learn coordinates from masked, keyed personality items, reconstruct complementary questionnaire forms, inspect a local metric in standardized item space, and prepare the coordinates for a future behavior-generating interface. This is an adaptation to personality representation, not a claim of a novel manifold-learning algorithm.

Why not simply choose Isomap or an ordinary autoencoder?

Isomap is valuable for testing graph-based structure, but does not by itself supply this learned generative decoder. An ordinary nonlinear autoencoder is not an opposing alternative: it already provides the encoder-decoder basis. The selected manifold model adds explicit decoder-based geometry. Neither a variational posterior nor a contrastive objective is logically required for that addition.

4. Mathematical and computational specification

Observation model and coordinates

Let x be a 300-item response vector and m its observation mask. Training-only means and standard deviations define standardized observed values; missing values are filled with zero after standardization and explicitly marked by m. The encoder receives [x, m], a 600-dimensional input, and produces a Gaussian posterior over coordinates z. In inference and evaluation, the posterior mean is used as the point representation.

$$q_\theta(z \mid x, m) = \mathcal{N}\big(\mu_\theta(x, m), \operatorname{diag}(\sigma^2_\theta(x, m))\big) \tag{1}$$

$$z = \mu_\theta(x, m) \in \mathbb{R}^{16}, \qquad \hat{x} = g_\psi(z) \in \mathbb{R}^{300} \tag{2}$$

Encoder: 600 -> 256 (SiLU) -> 128 (SiLU) -> posterior parameters. Decoder: 16 -> 128 (SiLU) -> 256 (SiLU) -> 300 predictions.

Learning objective

The implemented objective combines masked reconstruction, complementary-view reconstruction, a small KL term and an optional matching loss:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + 0.5\mathcal{L}_{\text{cross}} + 0.001\mathcal{L}_{\text{KL}} + 0.02\mathcal{L}_{\text{con}} \tag{3}$$

Cross-view loss averages the two view-reconstruction losses. KL is averaged per coordinate; the no-contrastive condition sets the last term to zero.

Input dropout is 0.3. Missing target items are excluded from neural reconstruction losses. The optional contrastive objective matches the two forms of the same training response case using symmetric InfoNCE, batch size 32 and temperature 0.2. The forms supply paired observations, not temporal sequences or independent personality labels.

Decoder-induced metric

$$J_g(z) = \frac{\partial g_\psi(z)}{\partial z}, \qquad G(z) = J_g(z)^\top J_g(z) + 10^{-4} I_{16} \tag{4}$$

$$L(c) = \int_0^1 \sqrt{\dot{c}(t)^\top G(c(t))\, \dot{c}(t)}\, \mathrm{d}t \tag{5}$$

The ridge makes the metric positive definite numerically, but does not establish that the decoder is everywhere full rank or globally injective. The observation-space metric is Euclidean in train-standardized item units. Alternative weighting or ordinal likelihoods would change the interpretation and should be investigated.

Current distance implementation is not a shortest geodesic

The current code estimates length along a straight line in latent coordinates using decoded finite segments and a ridge contribution. Training uses four segments; matching evaluation uses eight. This is a path-length proxy, not a solved shortest-path problem. A proper geodesic implementation would optimize interior path points under G, check convergence, and compare against straight-path lengths.

5. Data and experimental protocol

Primary evidence: IPIP item-level personality data

The source is the Kajonius and Johnson supplementary archive [5], using its Johnson 2005 300-item response file and official item key. Item mapping uses the Full# field rather than spreadsheet row order. The supplied response file already contains reverse-key recoding, so applying reverse scoring a second time would be incorrect.

Processing decision	Implemented rule
Valid item values	Values 1-5 are observed; zero, nonnumeric and out-of-range values are missing.
Exploratory response screening	At least 270 valid items; at least 8 observed per facet; RNDSPLIT >= 0.5 and RNDCORR >= 0. These are study screens, not clinical cutoffs.
Retained cases	20,511 eligible; 32 exact response-vector duplicates removed; 20,479 retained. Missing-item fraction approximately 0.525%.
Fixed split	14,335 training; 3,071 development; 3,073 reserved test. Split by response case, not verified unique person.
Complementary forms	Two nonoverlapping 150-item forms, each containing 5 items per facet; fixed data seed 42.
Optimization	30 epochs, AdamW, learning rate 0.001, weight decay 0.0001, batch size 256; training seeds 42, 43 and 44.
Checkpoint selection	Lowest development hidden-form-B standardized MSE when encoding form A. The reserved test set has not been scored.

What the experiment actually measures

Hidden-item RMSE and item R-squared measure recovery of form B from form A. Facet/domain R-squared uses observed hidden-form responses aggregated to 30 facets and five domains. Matching tests whether a query form retrieves its complementary form among the first 256 development cases. Chance top-1 is 1/256, or 0.39%. This evaluates same-occasion split-form consistency, not personality stability over months.

Fairness and inference limitations

Scaling is fitted only on training cases. The fixed split and forms are shared across runs. Nevertheless, dimension changes parameter count, and the ridge baseline mean-imputes missing training targets whereas the neural model masks them. Three-seed standard deviations quantify training variability, not population confidence intervals. Repeated PCA results across seeds are not independent replications.

6. Results and what they justify

All results below are exploratory development-set measurements. They are used to choose an implementation setting and expose limitations, not as final confirmatory evidence.

Latent size	Hidden-item RMSE mean +/- seed SD	PCA + ridge RMSE	Facet R2	Path top-1	Metric condition*
8	0.93738 +/- 0.00057	0.94750	0.5790	81.51%	4.26
16	0.91035 +/- 0.00027	0.91914	0.6278	94.14%	5.77
32	0.89771 +/- 0.00029	0.90051	0.6471	96.22%	159.73

Table 1. Three training seeds per dimension. *Mean of per-seed median metric condition numbers, not a direct psychological validity measure. Raw 150-item form-A ridge obtains RMSE 0.88150, lower than every latent model above.

Why use 16 dimensions provisionally?

Moving from 8 to 16 dimensions reduces RMSE by approximately 2.88%; moving from 16 to 32 reduces it by another 1.39%. The 32D model has the best prediction among the tested latent settings, but much greater metric anisotropy and cross-seed variation in its condition number. We provisionally use 16D as a compact geometry-oriented engineering compromise. This is post hoc development selection, not an estimate of the intrinsic dimension of personality, and 32D should remain an alternative if prediction is the overriding goal.

Does geometry-aware contrastive training help?

16D training condition	RMSE	Facet R2	Coordinate top-1	Decoder-path top-1
Decoder-path contrastive	0.91035	0.62784	91.15%	94.14%
Euclidean contrastive	0.91065	0.62733	92.19%	94.14%
No contrastive objective	0.91181	0.62533	90.89%	94.40%

Table 2. Means across seeds 42, 43 and 44; architecture held fixed at 303,180 parameters. Decoder-derived metrics can be computed for all three conditions, regardless of their contrastive objective.

Decoder-path training reduces RMSE relative to Euclidean training by only 0.000308, approximately 0.034%. It does not improve mean decoder-path matching, and the no-contrastive condition is slightly higher on that metric. Therefore, the contrastive geometry term is an optional experimental regularizer, not a demonstrated essential ingredient. The strongest current conclusion concerns the implementable decoder-based geometry, not a clear superiority of its training loss.

Posterior variance is close to its imposed lower bound: around 0.00034. Treat the current representation as near-deterministic; do not present its posterior variance as calibrated personality uncertainty.

7. How to form a digital personality with the manifold

Step 1 - Establish the evidence-grounded representation

Use the nonlinear encoder-decoder as the core representation method. Infer coordinates from observed personality evidence and its mask; retain provenance and measurement coverage. Decode the coordinates to inspect whether the representation reproduces plausible item-level patterns. Report coordinates as learned features rather than assigning unvalidated trait names to each axis.

Step 2 - Make geometry an explicit computational service

Compute the decoder Jacobian and local metric for each inferred point. Expose coordinate distance, decoder-path length and, after implementation, optimized geodesic distance as distinct quantities. Inspect eigenvalues and behavior under perturbation. Validate sensitivity to item weighting and scaling before interpreting distance as personality similarity.

Step 3 - Connect representation to behavior

Train a language model or behavioral policy conditioned on the personality coordinates together with task context. One option is a small conditioning network that converts coordinates into prefix vectors or intermediate-layer modulation. Another is joint training of a smaller generator and the representation model. A frozen LLM is only one prototyping option; it is not part of the mathematical definition of the method.

The generator needs personality-linked behavioral or conversational targets. Questionnaire reconstruction alone cannot teach how an agent should act in social situations. Avoid assuming that verbal prompts describing coordinates will faithfully realize the learned geometry. Hold out people and situations, then evaluate behavioral consistency, task quality and response diversity separately.

Step 4 - Learn persistence and change from longitudinal evidence

Separate stable person-level coordinates from fast situational states only when repeated observations can support that distinction. A future transition model can predict state changes given prior state and context, with stable coordinates conditioning those transitions. The density-distribution view [4] motivates the distinction; it does not prescribe this neural implementation. The current IPIP model has not tested temporal transitions.

Step 5 - Validate an operational personality interface

Test whether the same person has consistent coordinates across independent occasions; whether decoded changes correspond to externally measured changes; and whether a conditioned agent expresses the intended tendencies across unfamiliar situations. For controlled changes, distinguish a geometric path through model predictions from a psychologically or causally realistic developmental trajectory.

Concrete implementation conclusion: learn personality coordinates with a nonlinear encoder-decoder; define the manifold through the decoder and its induced metric; then condition a separately trained behavioral generator on these coordinates. Geometry should govern explicit similarity and path operations, not merely appear in a visualization.

8. Roadmap, conclusion and references

Priority	Next deliverable	Acceptance evidence
1. Representation	Freeze the selected 16D specification and audit item-level reconstructions.	Reproducible preprocessing; masked evaluation; decoder sensitivity and rank checks.
2. Geometry	Implement optimized paths and compare with straight-path proxies.	Convergence, nonincreasing path length, stable endpoints and decoded profiles.
3. Behavioral interface	Acquire linked personality/behavior data; train a coordinate-conditioned generator.	Generalization to held-out individuals and situations; personality-specific behavioral tests.
4. Confirmation	Lock analysis choices, then evaluate reserved data.	Uncertainty over cases/people as appropriate; comparison against simpler baselines.

Final conclusion

The selected method is a learned nonlinear personality representation with decoder-induced Riemannian geometry. It is chosen because it joins three necessary capabilities: compressing high-dimensional evidence, reconstructing that evidence, and supporting explicit geometric operations. Sixteen dimensions is a provisional compromise, not a scientific claim about how many dimensions personality possesses. Current experiments support feasibility and reveal limits; they do not establish predictive dominance, temporal stability or a completed digital agent.

Thus, the answer to HOW is architectural and operational: evidence -> nonlinear coordinates -> decoder-defined manifold and metric -> personality-conditioned behavior. The next decisive work is to implement the missing geometric and behavioral interfaces, rather than to run more reconstruction experiments and call their scores digital personality.

References

[1] Tenenbaum, J. B., de Silva, V., & Langford, J. C. (2000). A global geometric framework for nonlinear dimensionality reduction. Science, 290(5500), 2319-2323. Source.

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[3] Arvanitidis, G., Hansen, L. K., & Hauberg, S. (2018). Latent Space Oddity: on the curvature of deep generative models. ICLR. Source.

[4] Fleeson, W. (2001). Toward a structure- and process-integrated view of personality: Traits as density distributions of states. Journal of Personality and Social Psychology, 80(6), 1011-1027. Source.

[5] Kajonius, P. J., & Johnson, J. A. (2019). Supplementary materials to Assessing the structure of the Five Factor Model of Personality (IPIP-NEO-120) in the public domain. PsychArchives. Source.

Evidence note: Tables 1-2 summarize locally completed IPIP dimension-sweep and geometry-ablation experiments. They are primary project results, not published literature benchmarks. Literature supports the component mechanisms; the integrated digital-personality pathway is the methodological synthesis proposed here.